Prob. 2. (2 ÆT. Ch. 3, 1.1a P59)

Def-n. (Homogenous of degree n):
Suppose f is in $C^1(\mathbb{R}^2)$. We say that f is <u>homogen-s of degree</u> n for some integer $n \geq 1$, if f satisfies

(1) $$f(tx, ty) = t^n f(x, y).$$

Give examples of f'ns that are hom-s of degree 1, 2, 3. It's only necessary to give one ex. of each degree but be sure to show that your ex-s are indeed hom-s.

Sol-n.

By the def-n above, simple examples would be:

1. Homogenous of degree 1:

(2) $$f(x, y) = x + y$$

Note (i) $f(x,y) \in C^1(\mathbb{R}^2)$ (as it's poly-l) &

(ii) $f(tx, ty) = tx + ty = t(x+y) = t^1 f(x, y)$, $n = 1$.

2. Homogenous of degree 2:

(3) $$f(x, y) = x^2 + xy.$$

Note

(i) $f(x,y)$ in (3) $\in C^1(\mathbb{R}^2)$ (again pol-y-l) &

(ii) $f(tx, ty) = (tx)^2 + (tx)(ty) = t^2(x^2 + xy) = t^2 f(x, y)$, $n = 2$.

3. Homogenous of degree 3:

(4) $$f(x, y) = x^2 y + xy^2.$$ Note that

(i) $f(x, y) \in C^1(\mathbb{R}^2)$ (because f is a polynomial of 2 var-s) &

(ii) $f(tx, ty) = (tx)^2 \cdot (ty) + (tx) \cdot (ty)^2 = t^3(x^2 y + xy^2) = t^3 f(x, y) \Rightarrow n = 3.$

$\blacksquare$